

Colin Lin

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Experience

Stripe

Software Engineer

Toronto, ON

October 2024 - Present

- Built and scaled an internal AI agent platform powering **170K+ weekly executions**, automating merchant risk-review workflows and driving **\$4.25M+ annualized impact** (Python, Java, Langchain, LangGraph)
- Led development of an internal ChatGPT-like LLM chat platform with **10K+ monthly active users**, enabling MCP tool access, file/image uploads, and threaded conversations (React, TypeScript, Node.js, Go)
- Contributed to core infrastructure for a centralized **LLM gateway** handling **13B+ tokens/day** across **250+ use cases**, enabling model routing (OpenAI, Azure, AWS Bedrock), token/cost observability, and rate limiting
- Reduced LLM inference costs by **47% (\$5.6M/year)** by implementing Anthropic prompt caching with checkpoints across system prompts, tools, and conversation history to maximize cache hit rates in AI agent workloads
- Led resolution of a high-severity production incident in **74 minutes** impacting **100+ engineers** by diagnosing a cascading failure across client hooks, rate limiting, and Redis, and implementing fixes to prevent recurrence
- Reduced MongoDB latency from **9s to <1s** by profiling slow queries, validating bottlenecks via workload replay, and applying targeted index optimizations under production load

Konrad Group

Software Developer

Toronto, ON

September 2023 - August 2024

- Developed new frontend features for a major airline's flight booking application using **Vue.js**, impacting over **1M** users
- Built online contests and promotional pages for a major real estate client using **TypeScript** and **Next.js**
- Accelerated internal IT processes by **300%** by automating manual tasks using **Python** and **Selenium**
- Mitigated fraudulent contest entries by implementing browser fingerprinting using **JavaScript**

HSBC

Cloud Full Stack Engineer

Vancouver, BC

December 2021 - December 2022

- Developed cloud-based web apps for HSBC Wealth Compass and US Product Onboarding using **React** and **Redux**
- Engineered RESTful APIs using **Java Spring Boot** and **Express.js** microservices deployed on **AWS**
- Reduced maintenance cost by 40% by migrating legacy JavaScript codebase to **TypeScript**
- Collaborated with a cross-time-zone **Agile Scrum** team supporting Retail Banking systems across North America
- Mentored 3 new employees and interns on frontend development, API design, and advanced product knowledge

University of Toronto

Software Developer

Toronto, ON

September 2021 - February 2022

- Analysed trends across global news sites by constructing data visualizations using **Python**, **pandas**, and **Matplotlib**
- Improved the efficiency of the Python website crawler by 50% by applying filters and caching data
- Served browser-accessible **Jupyter Notebooks** for researchers using Jupyterhub deployed on the cloud

Education

University of Toronto

Honours Bachelor of Science in Computer Science

Toronto, ON

September 2019 - August 2023

- **GPA:** 3.88/4.0
- **Publications:** A Systematic Literature Mapping of COVID-19 Papers in Computer Science Education

Skills

Programming	Python, TypeScript, JavaScript, Go, Java, HTML, CSS, C, Swift, Kotlin, PHP
Frameworks and Libraries	React, Node.js, LangChain, LangGraph, Next.js, NumPy, pandas, Spring Boot, Angular, Vue.js
Databases	MongoDB, MySQL, PostgreSQL, Redis, Firestore, Neo4j
Tools	Git, Docker, Linux, AWS, Splunk, Databricks, Postman, nginx, Firebase, Jira